

Topological Board Game - Full Manual

Language: English

Topological Board Game is a board-game and research platform where familiar rules are played on unfamiliar spaces. Chess, Go, Reversi, Jump / Chinese Checkers, and Hex keep their original game identity, while topology, dimension, boundary conditions, lattice choices, robots, online rooms, and time-layer modes change the board on which those rules live.

1. Main Launcher

- Players enter normal game pages for quick play.
- Researchers enter Labs for reproducible discrete topological dynamics experiments.
- Life World is separate from Labs and focuses on cellular automata, artificial life, ecology, and species interaction.
- The globe icon switches English and Traditional Chinese.
- Home returns to the main launcher from game pages.

2. Common Interface

- Game board: the central interactive area.
- Game Controls: mode, board size, topology, boundary, lattice, robot, online, and new-game controls.
- Status card: current turn, winner, counts, clocks, warnings, or connection state.
- Move history: past moves and, in +1D modes, pending or replayed events.
- Chat panel: available in online pages.
- Camera controls: rotate, zoom, slice, focus, and reset options where available.

3. Input Controls

- 2D boards: click or tap a valid cell, intersection, or piece.
- 3D boards: rotate the board, then click the visible target site or face.
- 4D boards: select slices or projected layers, then choose a visible site.
- Mobile: tap controls; pinch or drag on supported boards.
- If a click seems blocked, rotate or zoom so the intended site is visible.

4. Board Spaces

- Standard open board: edges stop at the board boundary.
- Cylinder: one pair of opposite edges wraps.
- Torus: both edge directions wrap.
- Mobius strip: one edge wraps with a flip.

- Klein bottle: one direction wraps normally and the other wraps with orientation reversal.
- Sphere: a closed surface board, sometimes with pole sites or geodesic-style links.
- Projective plane / RP-style boundary: opposite boundary points are identified with reversal when supported.
- R3 / voxel volume: a 3D board volume.
- T3 / periodic volume: R3 with periodic boundary conditions.
- RBC / random boundary: a fixed random boundary connection map.

5. Lattices

- Square: the standard grid.
- Triangular: diagonal connections create triangular local cells.
- Honeycomb: hexagon-cell graph; Go usually plays on vertices, while Reversi and Hex may use faces depending on the page.
- BCC, FCC, HCP, and other 3D lattices: 3D neighbor structures where supported.
- Lattice choices change adjacency, legal paths, captures, flips, and connection tests.

6. Chess Rules

- Objective: checkmate the opponent king.
- King: moves one step in any legal adjacent direction and may not move into check.
- Queen: moves along legal straight lines.
- Rook: moves along orthogonal legal lines.
- Bishop: moves along diagonal legal lines.
- Knight: moves in the page's legal knight pattern.
- Pawn: advances according to the selected chess page rules and captures diagonally where supported.
- A move is illegal if it leaves your king in check.
- In topological and higher-dimensional boards, the displayed graph or geometry determines which directions and connections are legal.

7. Go Rules

- Objective: control more territory after captures and scoring.
- Stones are placed on intersections or graph vertices.
- A liberty is an adjacent empty point on the board graph.
- A connected group with no liberties is captured.
- Suicide, ko, superko, pass, komi, and scoring follow the selected page settings.
- On non-square topological boards, liberties and territories are counted through graph connectivity, not through a flat picture.
- Final scoring should label black and white territory clearly and include captures, komi, and dead-stone handling where the page supports it.

8. Reversi Rules

- Objective: finish with more discs than the opponent.
- A legal move places a disc so that one or more connected lines of opponent discs are bracketed between the new disc and an existing friendly disc.
- Bracketed discs flip to the player who moved.
- If no legal move exists, the turn may pass according to page rules.
- On topological boards, flip lines follow the board graph and available directions.
- For surface boards, some pages place discs on faces rather than vertices; follow the board explanation shown beside the controls.

9. Jump / Chinese Checkers Rules

- Objective: move your pieces into the target camp.
- A normal move goes to an adjacent empty site.
- A jump crosses an occupied neighboring site and lands on the empty site beyond it when the path is legal.
- Chain jumps may continue when supported.
- Topology changes the graph, so legal jumps follow shown connections.
- Robot turns should animate step by step so players can see the path.

10. Hex Rules

- Players are Blue and Orange.
- A move fills one empty Hex unit.
- Blue connects its two target zones.
- Orange connects its two target zones.
- There is no flipping, capture, movement, or pass by default.
- Standard 2D Hex usually uses a honeycomb-style cell board.
- Other topological Hex boards use explicit target zones or seams, not imaginary physical edges.
- A win is detected by graph connectivity through the player's filled cells.

11. Robots

- Robot mode lets a local player face the computer.
- Difficulty levels may change search depth, evaluation, rollout count, or heuristic strength.
- Stronger robots may use more time.
- If a robot move feels too fast to understand, use pages that expose animation speed or replay history.

12. Online Play

- Use Find Match for matchmaking when supported.
- Use Room controls to create or join a room.
- Online play depends on Firebase configuration.
- Email/password accounts require Email/Password sign-in to be enabled in Firebase Authentication.
- If online mode is developing for a feature, the page should say so clearly.

13. +1D Time Modes

Future Mode

- A player chooses an action now.
- The action is scheduled for a future tick.
- The original board-game rule is checked when the action resolves.
- If legal at resolve time, it applies.
- If illegal at resolve time, it fails and should be shown in history.

Past Mode

- Moves normally resolve immediately.
- A recent resolved event can be rewritten inside a fixed rewrite window.
- The rewrite replaces an event, then the timeline replays.
- If validation fails, the rewrite is rejected with a clear conflict message.
- No branch timeline is shown in the current UI.

Past+Future Mode

- Players can schedule future actions and rewrite recent past actions.
- Later events are replayed and revalidated.
- Pending future actions may be cancelled or rescheduled before they resolve.

Age And Go Time Period

- Age mode and noise can modify supported +1D games except where disabled.
- Go Time Period is a Go-specific time-period mode and should not be mixed into other games.
- Age counters and period counters are separate UI concepts.

14. Life World

- Life World is for cellular automata, artificial life, evolution, ecology, and species interaction.
- Basic play: Start, Step, Reset, Random seed, Draw, Erase, Inspect, species selection, and board size.

- Research controls: geometry, lattice, dimension, rule preset, B/S custom rule, neighborhood, noise, age, mutation, max generations, pattern JSON, observables, phase scanning, topology comparison, and experiment notebook.
- R3 Life uses a full 3D voxel volume.
- T3 periodic and RBC are R3 boundary-condition choices, not separate Life geometries.

15. Labs

- Labs is a research platform for discrete topological dynamics.
- It separates physics, mathematics, information, topology, graph dynamics, and research export from game progression.
- Labs pages emphasize topology, state space, local rule, dynamics, observables, experiment configuration, results, reproducibility, and export.
- Labs does not contain biological Life World models.

16. Saving And Export

- Many pages support JSON export/import.
- Labs and Life World may export observables, snapshots, experiment configs, CSV time series, or notebook entries.
- Keep exported files if you need reproducibility across versions.

17. Performance Tips

- Reduce board size for slow 3D or 4D boards.
- Prefer simpler lattices for older devices.
- Use WebGL-capable browsers and updated GPU drivers.
- For research scans, Web Workers or offline Python tools can improve responsiveness.

18. Support

Use the README and Troubleshooting PDF for installation, account, online, and reset help. For online manuals, link the Steam Basic Info tab to the public Topoboardgame website manual page.